Fundamentals of Cybersecurity

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Introduction



Security

A simple definition:

- Protection of what matters to us



Computer security

- Protection of items you value, called the **assets** of a computer or computer system (from harm, loss, or unauthorized access)

The Vulnerability–Threat–Control paradigm

Vulnerability

- A weakness in the system—for example, in procedures, design, or implementation—that might be exploited to cause loss or harm.
- It does not cause harm on its own but creates the potential for a threat to occur.





The Vulnerability–Threat–Control paradigm

Threat

- A potential event or action that could exploit a vulnerability and cause harm or loss.
- It can be deliberate or accidental.





The Vulnerability–Threat–Control Paradigm

How do we address these problems?

- We use a **control** or **countermeasure** as protection.
- A control is an action, device, procedure, or technique that removes or reduces a vulnerability.





A **threat** is blocked by **control** of a **vulnerability**.

CIA triad

- Confidentiality
  - The ability of a system to ensure that an asset is **viewed** only by authorized parties.
- Integrity
  - The ability of a system to ensure that an asset is **modified** only by authorized parties.
- Availability
  - The ability of a system to ensure that an asset can be **used** by any authorized parties.

Three main properties of security

- Authentication: the ability of a system to confirm the identity of a sender.
- Non-repudiation or accountability: the ability of a system to confirm that a sender cannot convincingly deny having sent something.

Types of threats

- Harm can be caused by either **humans** or **nonhuman threats**.
- In this course, we focus primarily on human threats.
  - Non-malicious
  - Malicious
  - Attack: a **malicious person** wants to cause harm



Cyberthreats

- **Computer security** often refers to individual computing devices
- However, these devices seldom stand alone; they are usually connected to networks of other computers.
- Enlarging the scope from one device to many devices, users, and connections leads to the word "cyber."
  - Cyberthreat: a threat not just against a single computer but against many computers that belong to a network.
  - Cyberspace: the online world of computers, especially the Internet.
  - Cybercrime: an illegal attack against computers connected to network, as well as their users, data, services, and infrastructure.
- We examine security as it applies to individual computing devices as well as devices in networks, that is, cybersecurity.

Controls

- A **control** or countermeasure is a means to counter threats.
- Harm occurs when a **threat** is realized against a **vulnerability**.
- We can deal with harm in several ways:
  - *prevent* it, by blocking the attack or closing the vulnerability
  - *deter* it, by making the attack harder but not impossible
  - *deflect* it, by making another target more attractive (or this one less so)
  - *mitigate* it, by making its impact less severe
  - *detect* it, either as it happens or some time after the fact
  - *recover* from its effects

What's next?

- Chapter 2: three powerful and widely used security protection methods: identification and authentication, access control, and encryption. We call these three our security toolkit.
- Chapter 3: problems and protections of computer code. We also consider malicious code, such as viruses and Trojan horses.
- Chapter 4: how networked computing affects individuals, primarily through browsers and other basic network interactions such as email.
- Chapter 5: protections operating systems can provide to users' programs and data.
- Chapter 6: looking at the whole network and its impact on users' abilities to communicate data securely across the network.
- Chapter 7: data, databases, and data mining.
- Chapter 8: cloud computing and Internet of Things.
- Chapter 9: privacy

Course references

- Main reference
  - Charles P. Pfleeger, ***Security in Computing***, 6th edition (2024)
- Other useful textbooks
  - William Stallings, ***Network Security Essentials: Applications and Standards***, 6th edition (2017).
  - William Stallings, **Cryptography and Network Security Principles and Practice**, 8th edition (2023).

Evaluation

- Assignments: 6
- Mid-term exam: 6
- Final exam: 8
- Active participation in class: 2